TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

# Electrical Machines: D.C. Motor Speed Control, Synchronous Motor and Synchronous Generator

*A d.c. motor's speed answers to two handles — the voltage on its armature and the flux in its poles. The synchronous machines, by contrast, are locked to the supply frequency, and their behaviour is read from two famous curves.*

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## 1. Aim and theory

**Aim:** (i) to control the speed of a d.c. shunt motor by the armature-voltage and field-flux methods and to draw the speed characteristics; (ii) to study the construction, starting and V-curves of the synchronous motor; (iii) to study the synchronous generator and to plot its open-circuit characteristic (OCC).

**D.C. motor.** The rotating armature generates a back e.m.f. $E_b = V - I_a R_a$ opposing the supply, and the speed is set by

$$N \propto E_b / \Phi \approx (V - I_a R_a) / \Phi$$

Two practical handles follow. **Armature-voltage control** varies V (variac-fed rectifier, chopper, or a Ward–Leonard motor–generator set) with the field at full strength: speed changes smoothly *below* base speed, torque capability stays constant, and no power is wasted in rheostats. **Field-flux control** weakens Φ with a rheostat in the field circuit: speed rises *above* base speed, but torque falls as flux falls, so the drive is roughly constant-power. (A third, older method — a resistance in the armature circuit — is wasteful and gives poor speed regulation; it survives mainly in starters.)

**Synchronous motor.** The stator carries a three-phase winding producing a field rotating at $N_s = 120f/P$; the rotor (salient-pole at low speeds, cylindrical at high) carries d.c.-excited poles fed through slip rings or a brushless exciter. At standstill the stator field sweeps past the rotor poles so fast that the torque alternates and averages zero — the motor is **not self-starting**. It is brought up to near $N_s$ by its damper (amortisseur) winding acting as a squirrel cage, by a pony motor, or by a variable-frequency drive; the d.c. excitation is then applied and the rotor locks into step. Once locked, it runs at exactly $N_s$ whatever the load (within pull-out torque). Varying the field current at constant load traces the **V-curves**:

armature current against field current is V-shaped, minimum at unity power factor; under-excited the motor draws lagging current, over-excited it draws leading current — the property used when an over-excited synchronous motor is run as a *synchronous condenser* to correct a factory's power factor.

**Synchronous generator (alternator).** The same machine run as a generator: a prime mover turns the excited rotor and the stator winding delivers three-phase power, $E_{ph} = 4.44 \cdot f \cdot \Phi \cdot T_{ph} \cdot K_w$ per phase. On open circuit at constant speed, the terminal voltage plotted against field current gives the **OCC**: nearly straight at first (the unsaturated *air-gap line*), then bending over as the iron saturates.

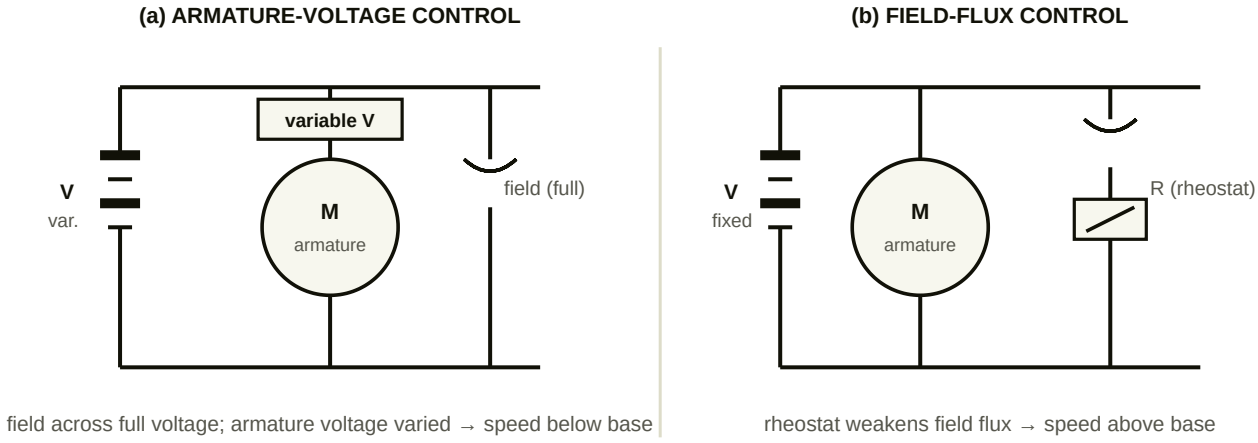



Figure 1 — The two speed-control connections of a d.c. shunt motor. (a) The field winding sits across the full supply; a variable voltage feeds the armature. (b) The armature sits across the fixed supply; a rheostat in series with the field winding trims the flux. The field must never be left open-circuited in either arrangement.

## 2. Apparatus and procedure

- **Apparatus:** d.c. shunt motor with starter, field rheostat and a variable d.c. supply (or Ward–Leonard set), tachometer, voltmeters and ammeters; three-phase synchronous motor with d.c. excitation supply, wattmeters and power-factor meter; synchronous generator driven at constant speed (d.c. motor or engine drive) with field rheostat and voltmeter.
- **Armature-voltage control:** run the motor at full field. Step the armature voltage down from 220 V and record speed at each step on a fixed light load; plot N against V — a nearly straight line through the origin region.
- **Field control:** restore full armature voltage. Insert field resistance in steps, recording field current and speed; plot N against $I_f$ — speed climbs steeply as the field weakens. Stop well before the maker's maximum speed.
- **Synchronous motor study:** examine the stator, rotor, slip rings and damper bars. Start the machine on its damper winding, apply excitation near synchronism, then at constant mechanical load vary the field current both ways and record armature current and power factor — the V-curve.

- **Synchronous generator OCC:** run the alternator at rated speed with its terminals open. From zero, raise the field current in steps and record the line voltage; continue a little past rated voltage, then come back down the same steps to expose any hysteresis. Plot $V_{oc}$ against $I_f$.

| CONTROL | SETTING | SPEED (RPM) | NOTE |
|---|---|---|---|
| Armature voltage | 220 V | 1,000 | base speed, full flux |
| Armature voltage | 190 V | 857 | $N \propto E_b$: 180/210 of base |
| Armature voltage | 165 V | 738 | 155/210 of base |
| Field flux | $\Phi$ = 90 % | 1,111 | $N \propto 1/\Phi$ |
| Field flux | $\Phi$ = 80 % | 1,250 | torque capability falling |

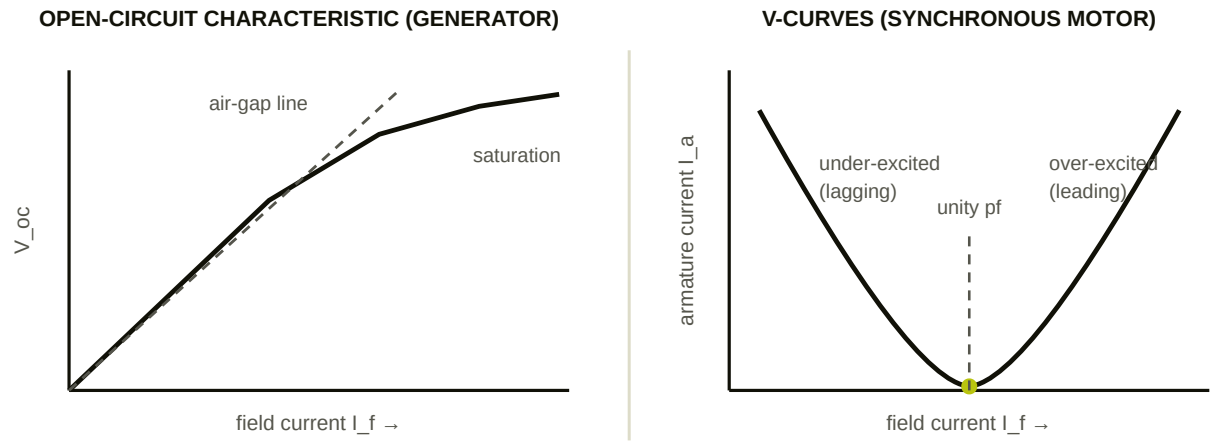



Figure 2 — The two signature curves. Left: the OCC follows the air-gap line until the iron saturates and the curve bends flat. Right: the V-curve of a synchronous motor at constant load — armature current is minimum where the power factor is unity.

## 3. Worked calculation

A 220 V d.c. shunt motor has $R_a = 0.5\ \Omega$ and draws $I_a = 20$ A at 1,000 rpm with full flux.

$$E_b = V - I_a R_a = 220 - 20 \times 0.5 = 210 \text{ V}$$

**Armature-voltage control:** reduce the armature supply to 165 V at full flux (armature current roughly unchanged at the same load torque, so $E_b' = 165 - 10 = 155$ V):

$$N' = 1{,}000 \times 155 / 210 = 738 \text{ rpm}$$

**Field control:** back at 220 V, weaken the flux to 90 % ($E_b$ stays ≈ 210 V at this estimate):

$$N'' = 1{,}000 \times (210 / 210) \times (1 / 0.90) = 1{,}111 \text{ rpm}$$

And for the alternator of the OCC test: a 4-pole machine driven at 1,500 rpm delivers $f = P \cdot N/120 = 4 \times 1{,}500/120 = 50$ Hz — the OCC must be taken at this constant speed, because the generated e.m.f. is proportional to both flux and speed.

## 4. Precautions

- **Never open the field circuit of a running shunt motor.** If $\Phi$ collapses while the armature is live, speed runs away toward destruction — the classic d.c. machine accident. Always keep a discharge path and reduce the armature voltage first when stopping.
- Start the d.c. motor on its starter so the armature current is limited while $E_b$ is still zero; never switch full voltage directly onto a stationary armature.
- During field weakening, watch the tachometer continuously and stay within the nameplate maximum speed — commutation also worsens at weak field.
- Apply d.c. excitation to the synchronous motor only near synchronous speed; exciting a stationary or slow rotor gives violent pulsating torque and heavy currents.
- Hold the alternator's speed steady during the OCC; a drifting speed shifts every voltage reading, since $E \propto N \cdot \Phi$.

## Practice (viva preparation)

1. Why does armature-voltage control work only below base speed, and field control only above it?
2. A shunt motor runs at 1,200 rpm with $E_b$ = 200 V. The flux is weakened by 20 % with $E_b$ unchanged. Find the new speed.
3. Why is a plain synchronous motor not self-starting, and how does the damper winding start it?
4. What is a V-curve, and what is special about its lowest point?
5. Why does the OCC bend over at high field current while the air-gap line stays straight?

### Answer key

1. Armature voltage cannot exceed the rated voltage (insulation and commutation limits), so it can only be reduced from rated — hence speeds below base. Weakening the flux below its rated value raises speed, but the flux cannot usefully be strengthened beyond rated (saturation), so field control operates above base speed.
2. $N' = 1{,}200 \times 1/0.80 = 1{,}500$ rpm.
3. At standstill the rotating stator field sweeps past the stationary rotor poles at synchronous speed, so the torque reverses every half-cycle and averages to zero. The damper (squirrel-cage) bars in the

pole faces develop induction-motor torque and accelerate the rotor to near $N_s$; d.c. excitation is then applied and the rotor pulls into step.

4. A plot of armature current against field current at constant mechanical load; V-shaped. At its minimum the motor takes its smallest current and works at unity power factor — under-excited on one side (lagging), over-excited on the other (leading).
5. The air-gap line is the voltage the machine would generate if the iron needed no m.m.f. (no saturation). In the real machine, rising flux drives the iron into saturation, so each extra ampere of field current buys less flux and less voltage — the OCC bends away from the air-gap line.

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